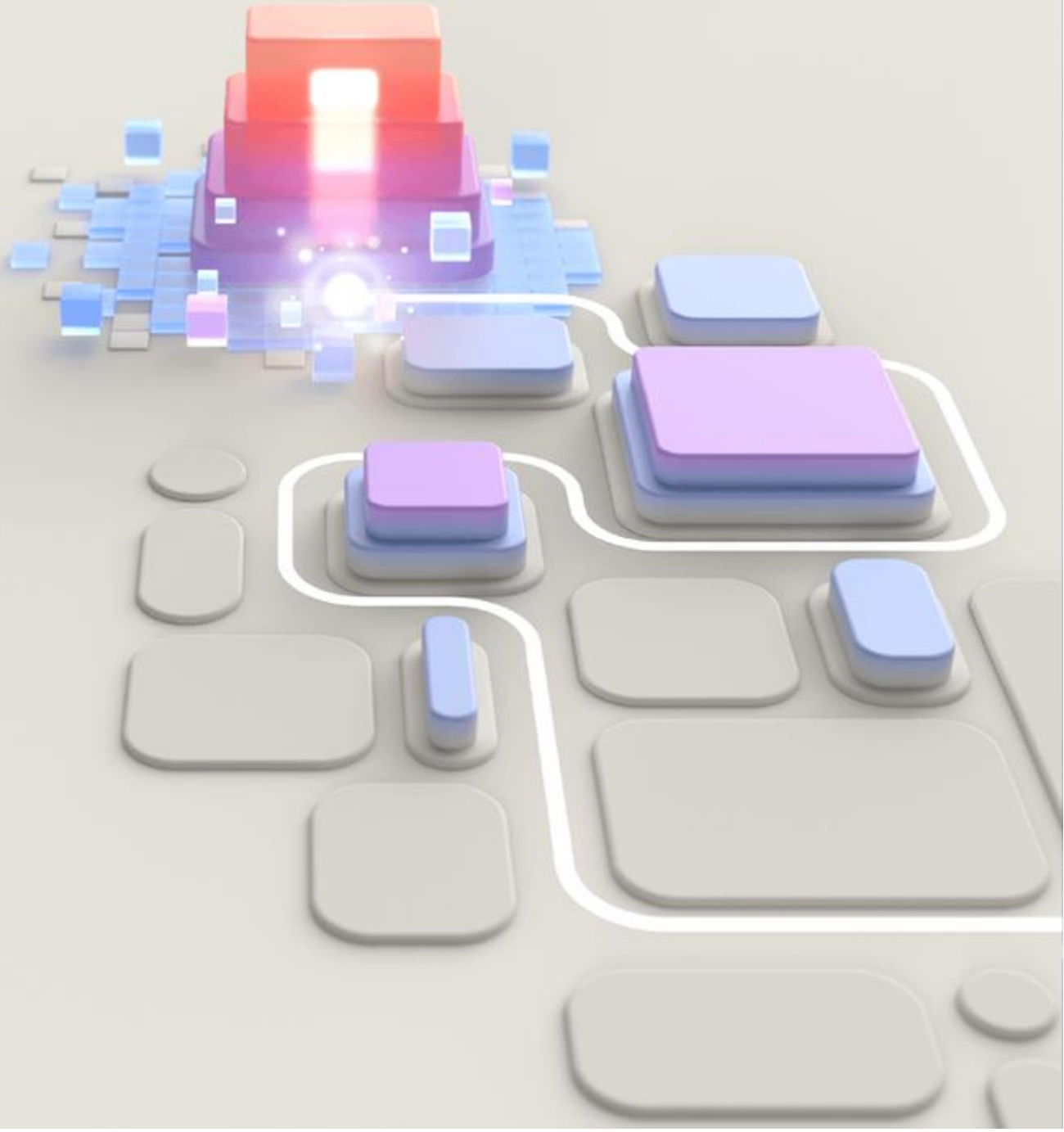
Microsoft

AZ-104

Administer Monitoring



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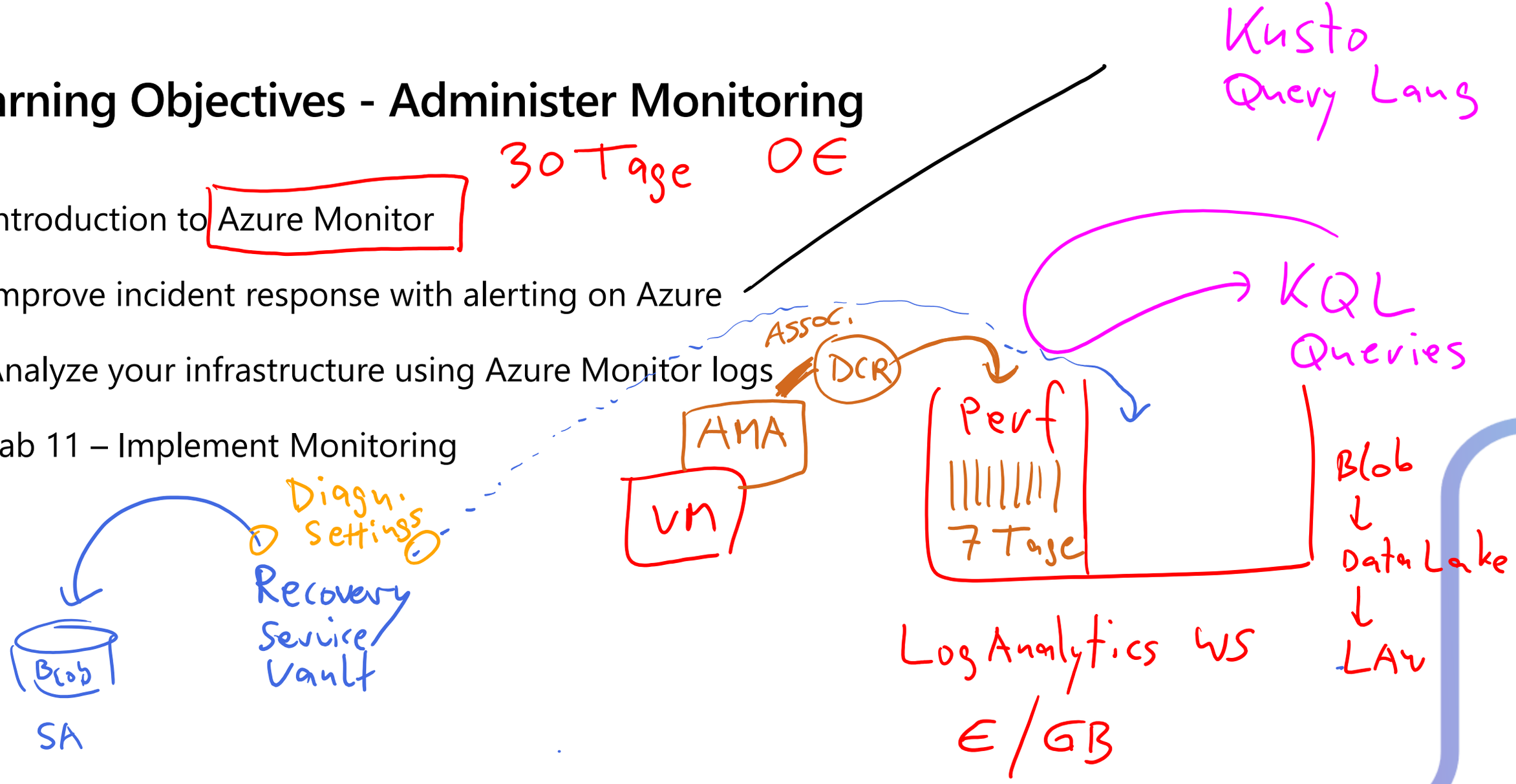
AZ-104 Agenda

01: Administer Identity
02: Administer Governance and Compliance
03: Administer Azure Resources
04: Administer Virtual Networking
05: Administer Intersite Connectivity
06: Administer Network Traffic Management
07: Administer Azure Storage
08: Administer Azure Virtual Machines
09: Administer PaaS Compute Options
10: Administer Data Protection
11: Administer Monitoring

Learning Objectives - Administer Monitoring

- Introduction to Azure Monitor
- Improve incident response with alerting on Azure
- Analyze your infrastructure using Azure Monitor logs
- Lab 11 – Implement Monitoring



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Introduction to Azure Monitor

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Learning Objective – Introduction to Azure Monitor

- Describe Azure Monitor Key Capabilities
- Describe Azure Monitor Components
- Define Metrics and Logs
- Identify Data Types
- Describe Activity Log Events
- Learning Recap

Monitor resources in Azure

- Interpret metrics in Azure Monitor
- Configure log settings in Azure Monitor
- Configure and interpret monitoring of virtual machines, storage accounts, and networks by using Azure Monitor Insights

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Describe Azure Monitor Key Capabilities

Monitor & Visualize Metrics

Metrics are numerical values available from Azure Resources helping you understand the health, operation & performance of your systems.

Explore Metrics

Query & Analyze Logs

Logs are activity logs, diagnostic logs and telemetry from monitoring solutions; Analytics queries help with troubleshooting & visualizations.

Search Logs

Setup Alert & Actions

Alerts notify you of critical conditions and potentially take corrective automated actions based on triggers from metrics or logs.

Create Alert

Core monitoring for Azure services

Collects metrics, activity logs, and diagnostic logs

Use for time critical alerts and notifications

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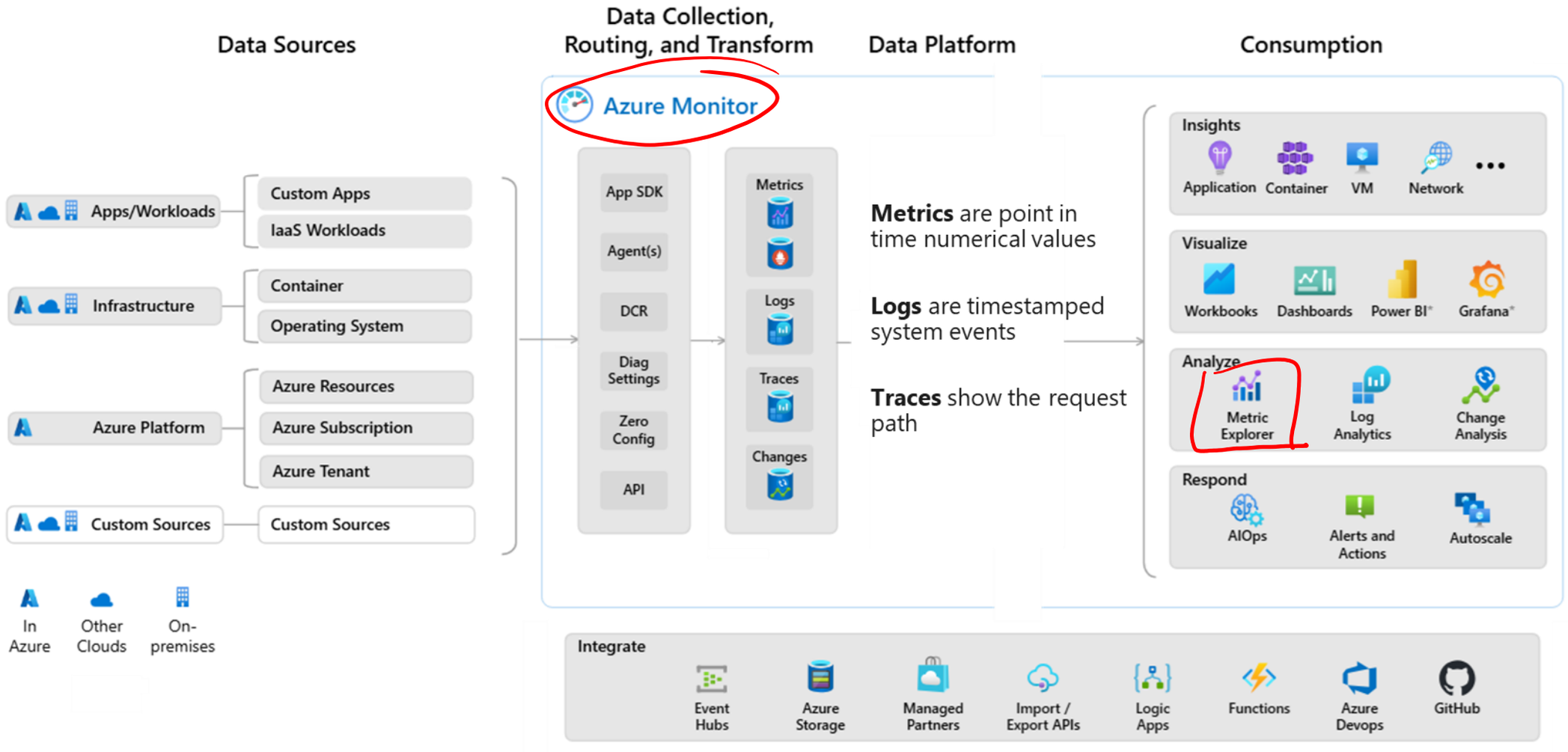
Understand Azure Monitor Components



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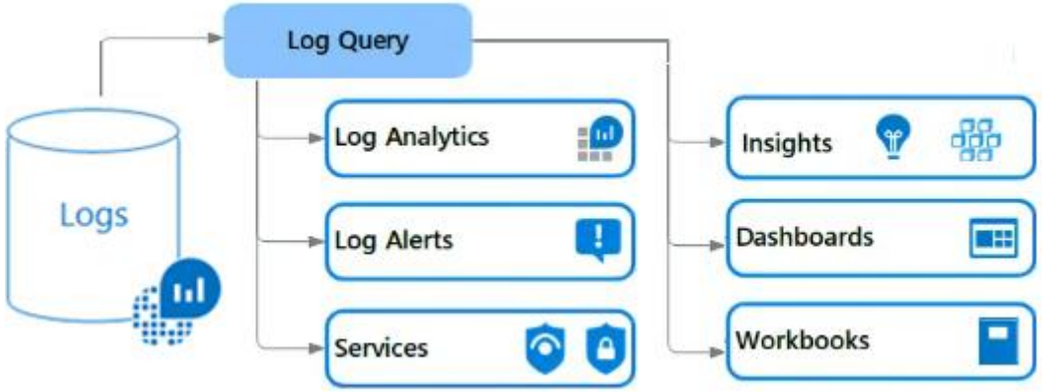
Define Metrics and Logs



- Metrics are numerical values that describe some aspect of a system at a point in time
- They are lightweight and capable of supporting near real-time scenarios

- Logs contain different kinds of data organized into records with different sets of properties for each type
- Telemetry (events, traces) and performance data can be combined for analysis

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Describe Activity Log Events

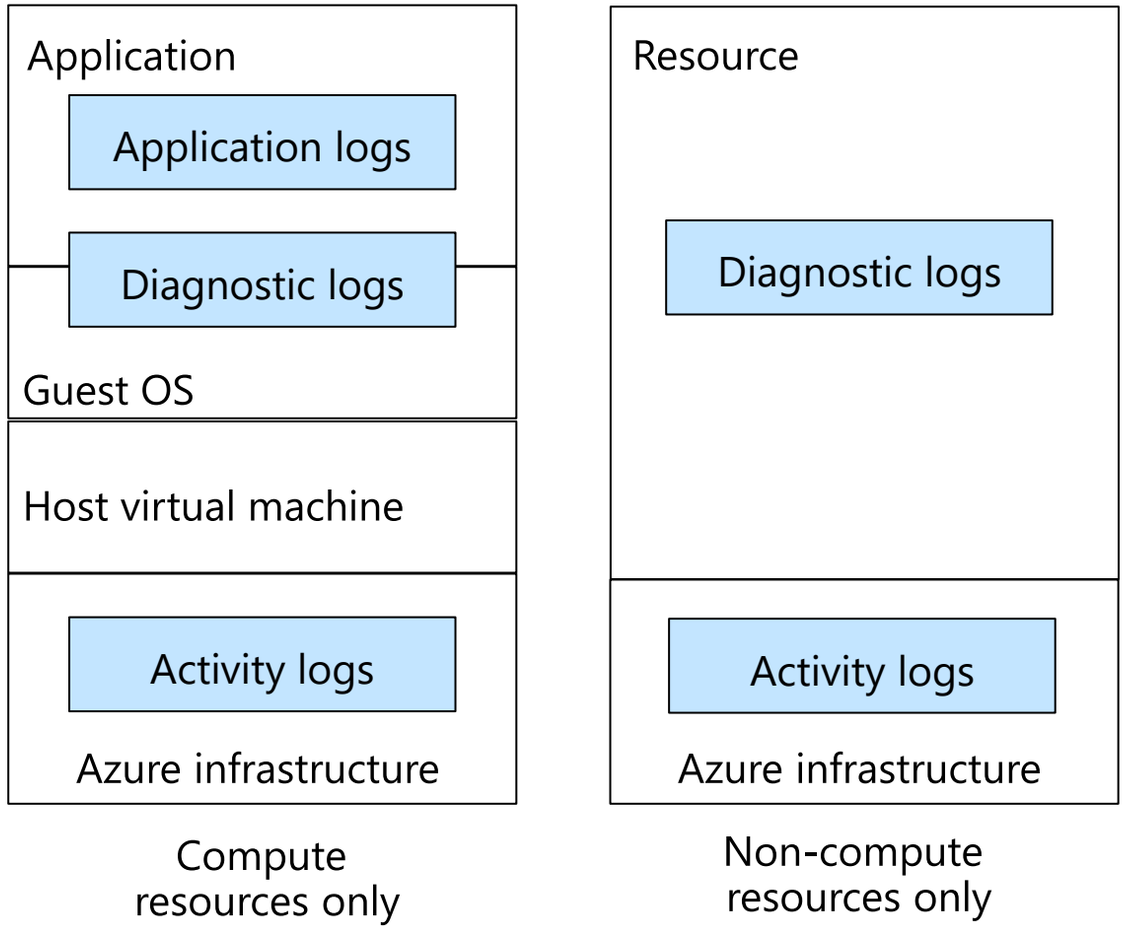
Send data to Log Analytics for advanced search and alerts

Query or manage events in the Portal, PowerShell, CLI, and REST API

Stream information to Event Hub

Archive data to a storage account

Analyze data with Power BI



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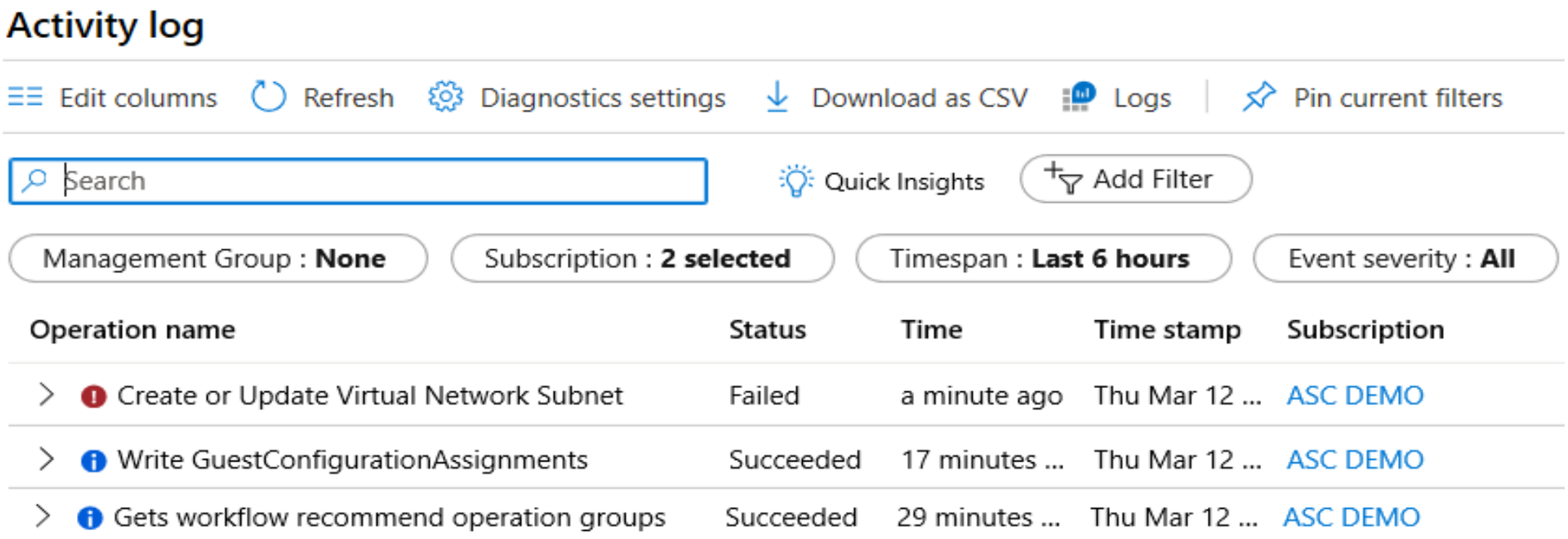
Query the Activity Log



Filter by Management group, Subscription, Timespan, and Event Severity

Add a filter, like Event Category (Security, Recommendations, Alerts)

Pin current filters and download as CSV

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Improve incident response with alerting on Azure

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Learning Objective - Improve incident response with alerting

- Manage Azure Monitor Alerts
- Create Alert Rules
- Create Action Groups
- Demonstration – Alerts
- Learning Recap

Monitor resources in Azure

- Set up alert rules, action groups, and alert processing rules in Azure Monitor

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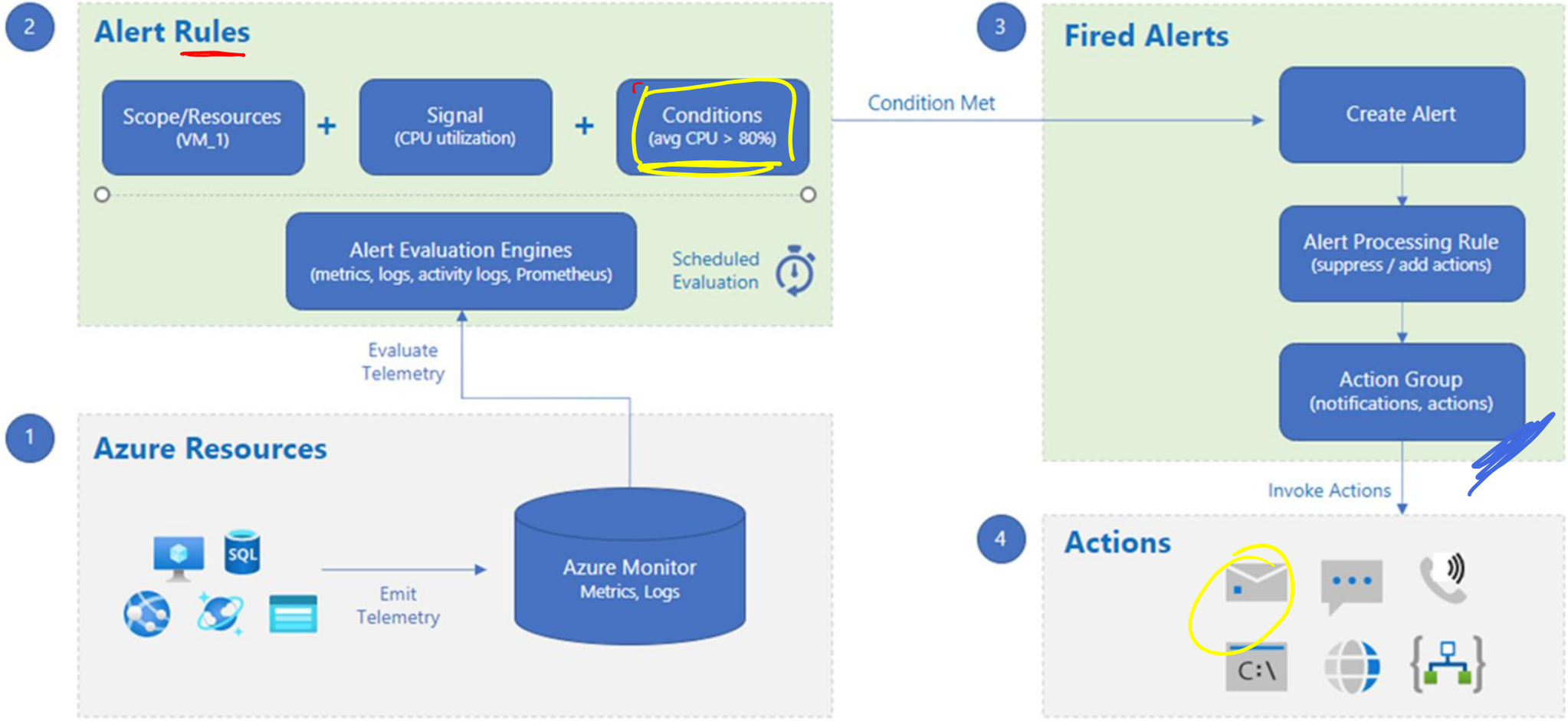
Manage Azure Monitor Alerts



Event Grid

wf

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Create Alert Rules

Home > Monitor | Alerts >

Alert rules ...

Name	Condition	Severity	Target scope	Target resource type	Signal type	Status
AzureSecurityCenter	Table rows > 1	4 - Verbose	export2LogA	Log Analytics workspace	Log search	Enabled
CPU Usage Percentage	node_cpu_usage_percentage > 80	3 - Informational	Demo	Kubernetes service	Metrics	Enabled
Failure Anomalies - HumanResources	Failure Anomalies detected	3 - Informational	humanresources	Application Insights	Smart detector	Enabled

- Alert rules combine the resources to be monitored, the signal or data from the resource, and the conditions.
- You can enable recommended out-of-the-box alert rules in the Azure portal.

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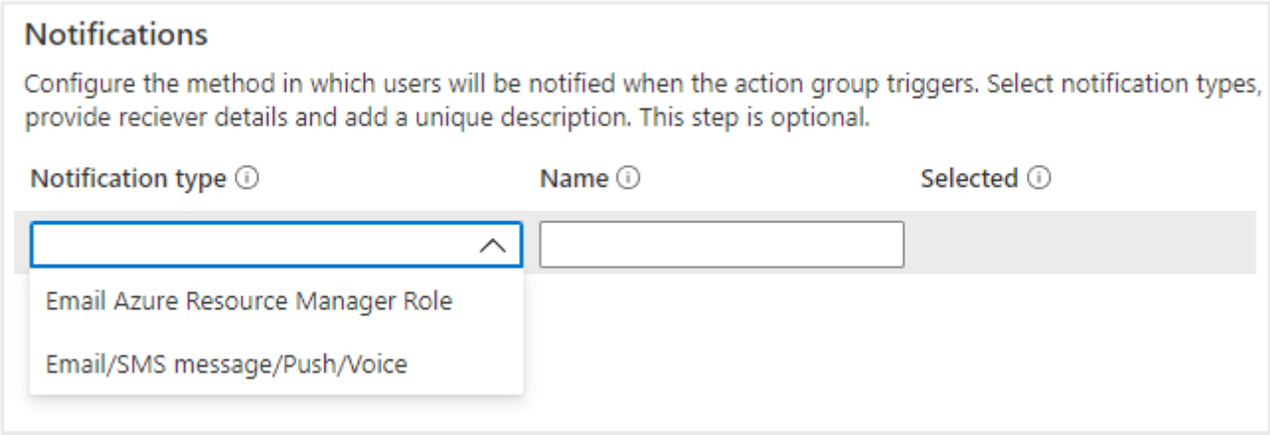
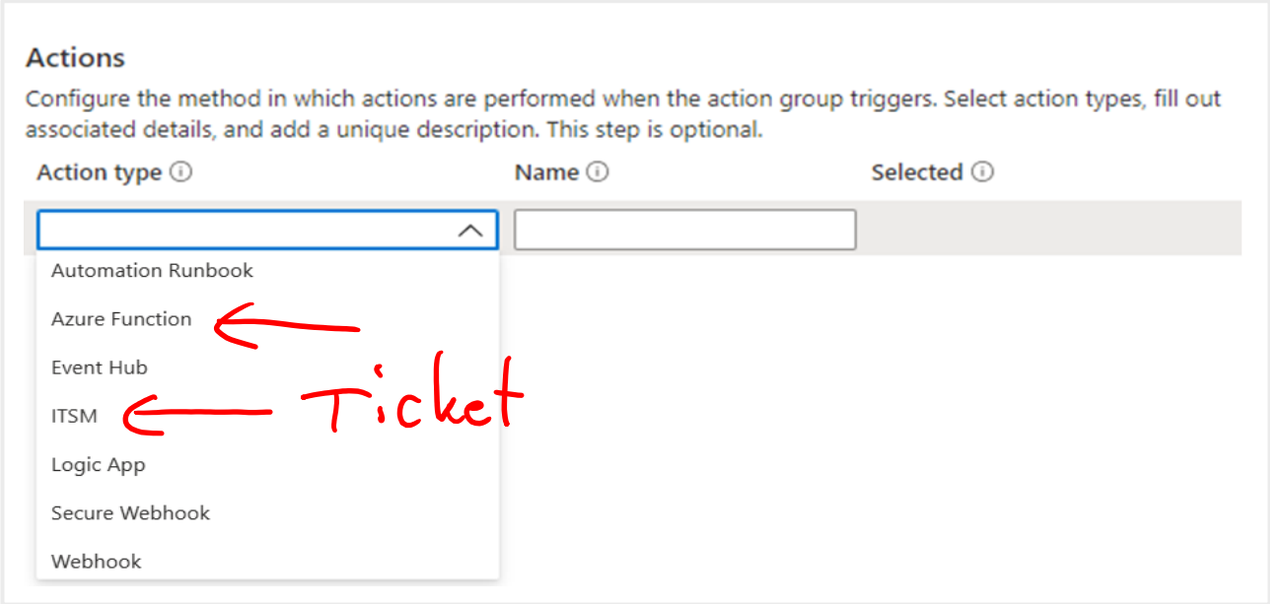
Create Action Groups

Defines a set of notifications and/or actions when an alert is triggered

You can add up to five action groups to an alert rule. Multiple alert rules can use the same action group.





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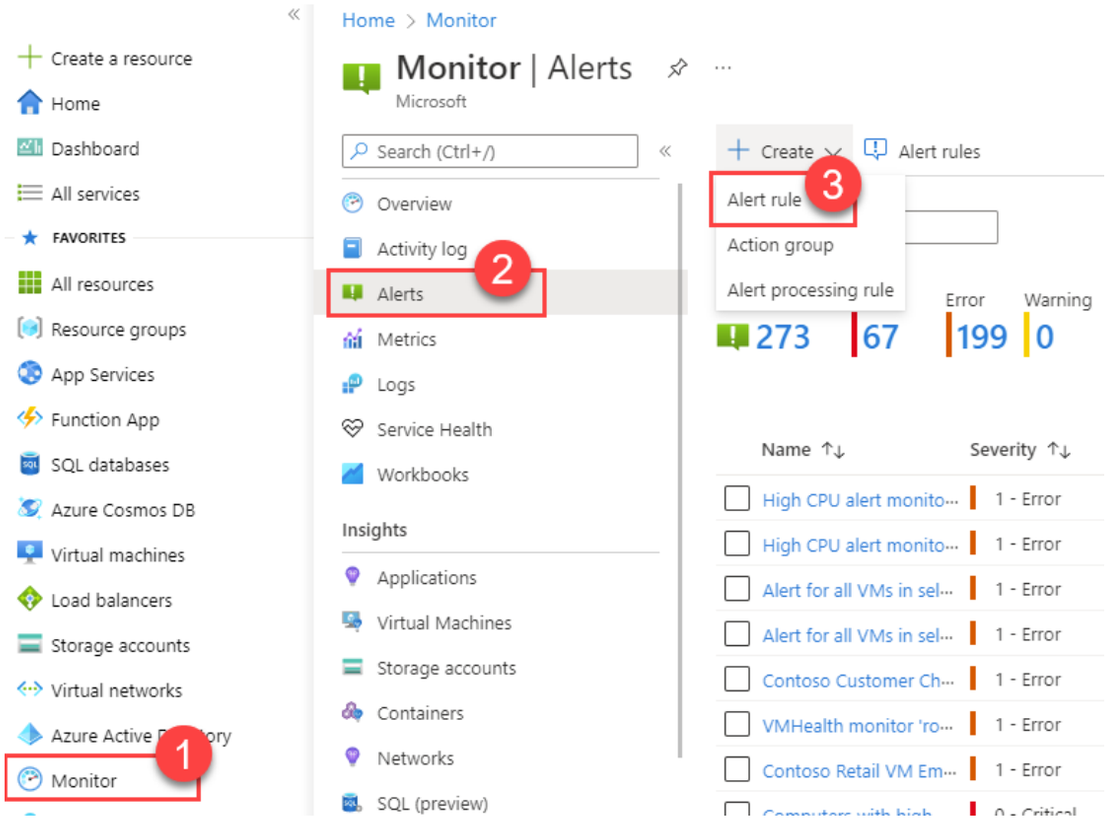
Demonstration – Alerts

- Create and configure an alert rule
- Review alerts



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Analyze your infrastructure using Azure Monitor logs

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Learning Objective - Analyze your infrastructure using Azure Monitor logs

- Determine Log Analytics Uses
- Create a Workspace
- Query Log Analytics Data
- Structure Log Analytics Queries
- Demonstration – Log Analytics
- Learning Recap

Monitor and maintain Azure resources (10–15%): Monitor resources in Azure

- Query and analyze logs in Azure Monitor

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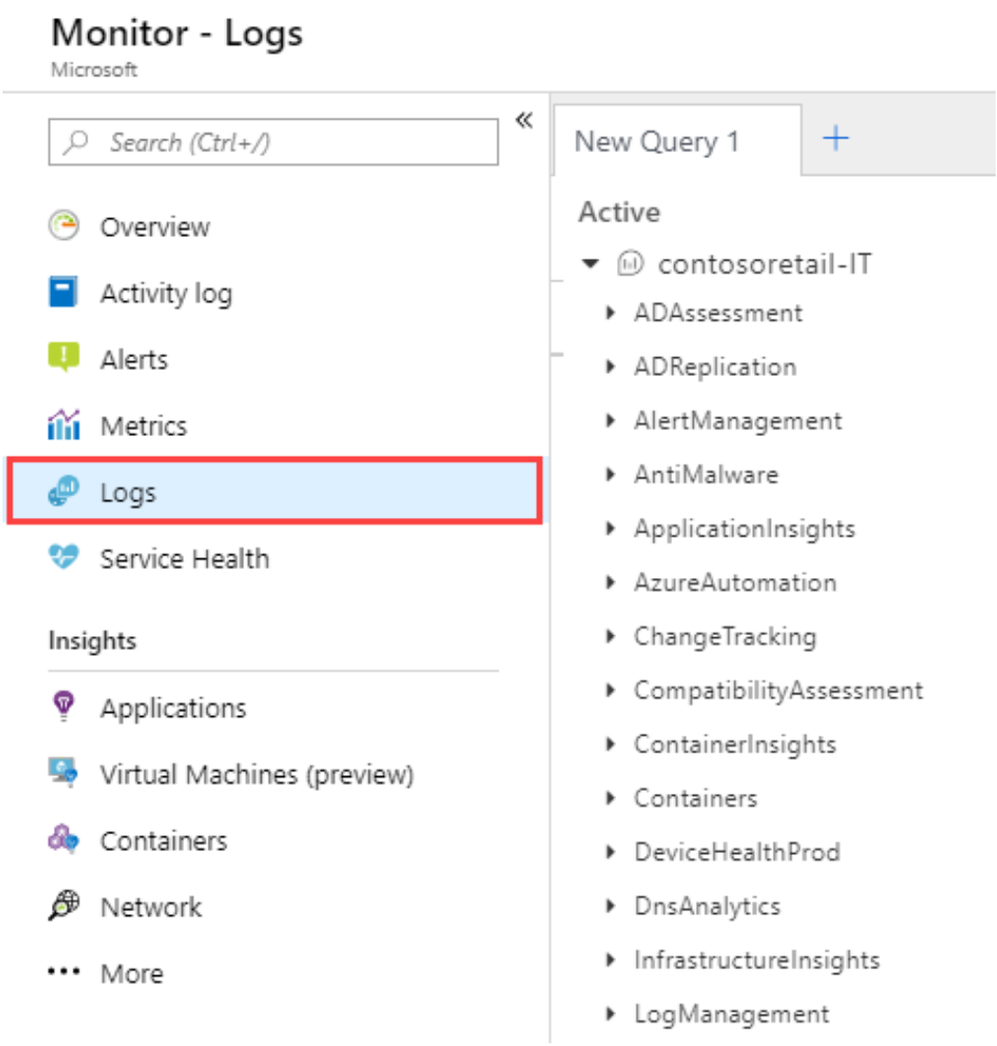
Determine Log Analytics Uses

A service that helps you collect and analyze data generated by resources in your cloud and on-premises environments

Write log queries and interactively analyze their results

Examples include assessing system updates and troubleshooting operational incidents



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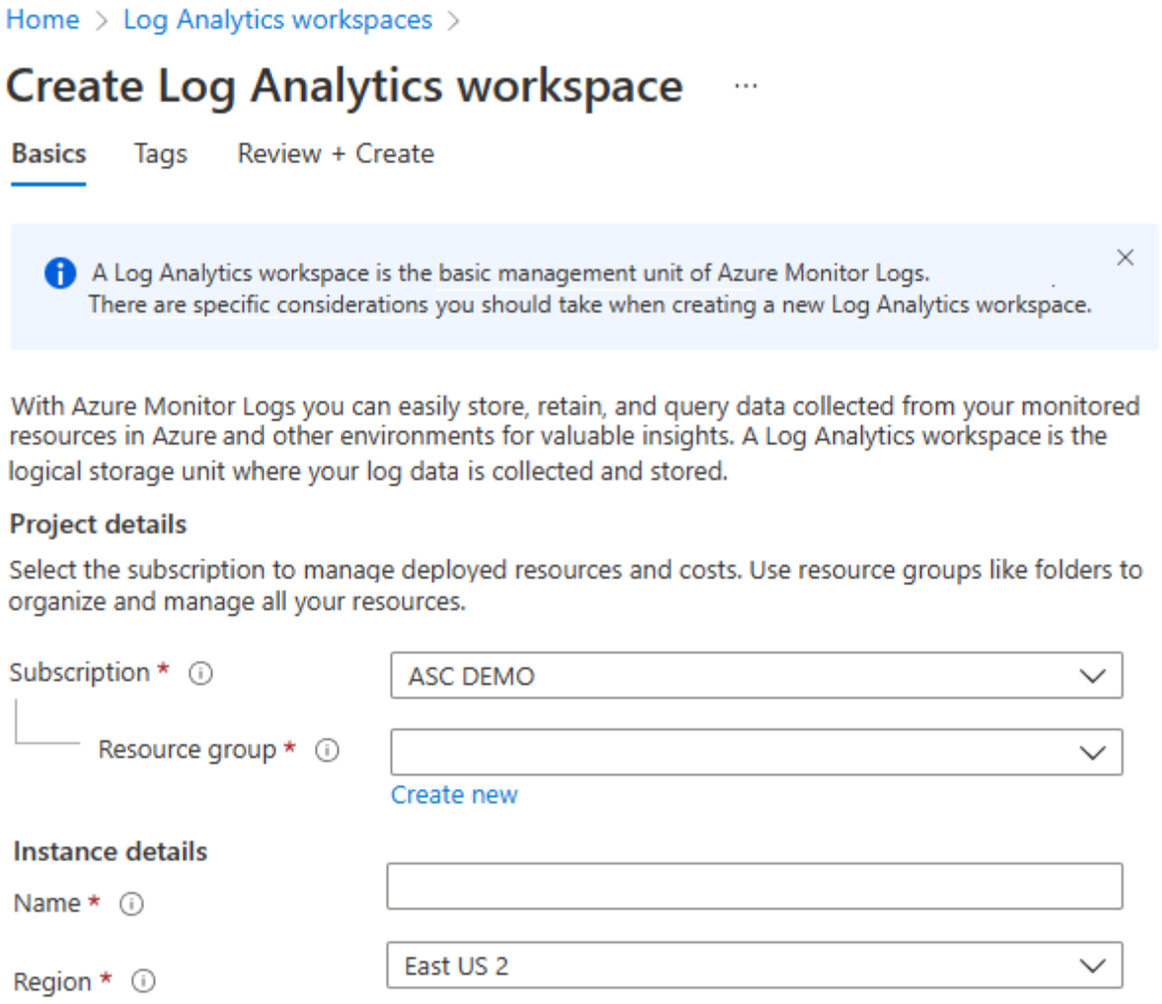
Create a Workspace

A workspace is an Azure resource and is a container where data is collected, aggregated, analyzed, and presented

You can have multiple workspaces per Azure subscription, and you can have access to more than one workspace

A workspace provides a geographic location, data isolation, and scope



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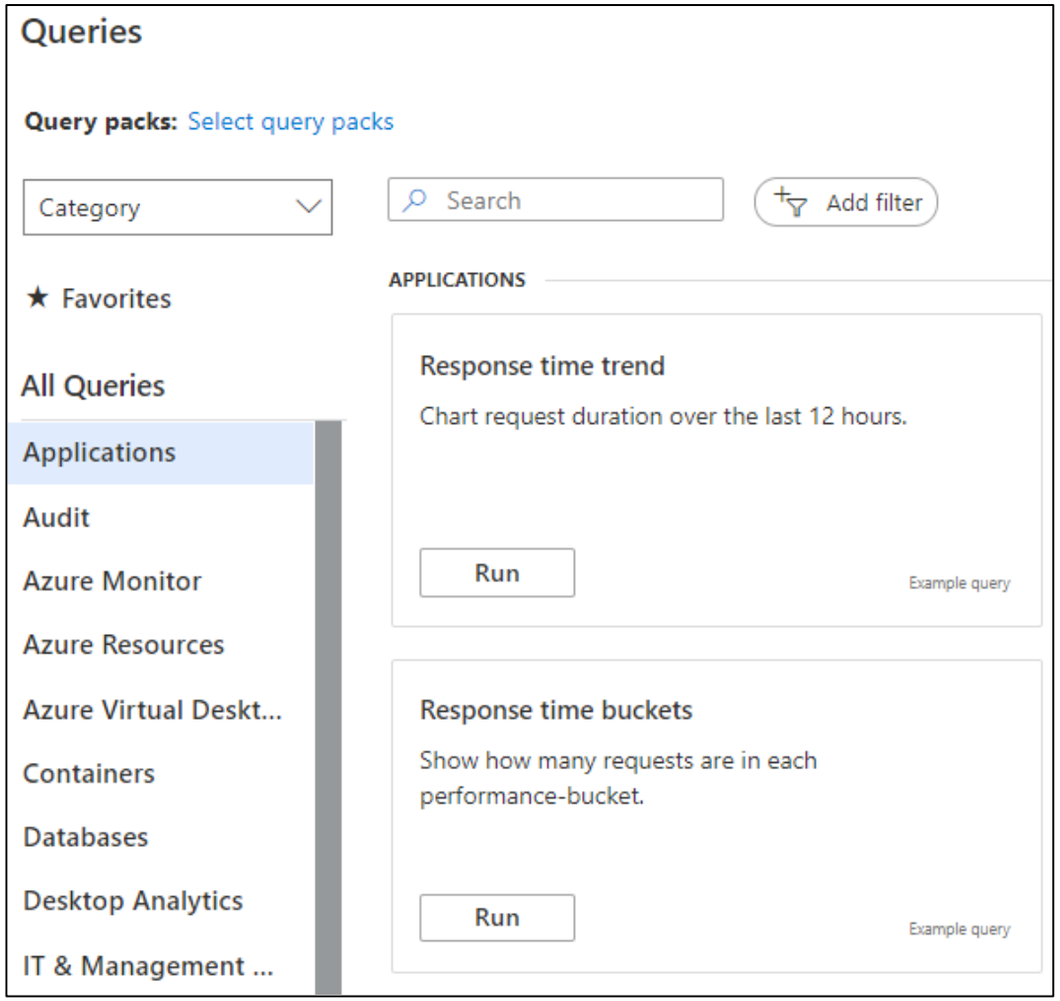
Query Log Analytics Data

Common queries and a query language (KQL) for custom searches

Quickly retrieve and consolidate data in the repository

Save or have log searches run automatically to create an alert

Export the data to Power BI or Excel



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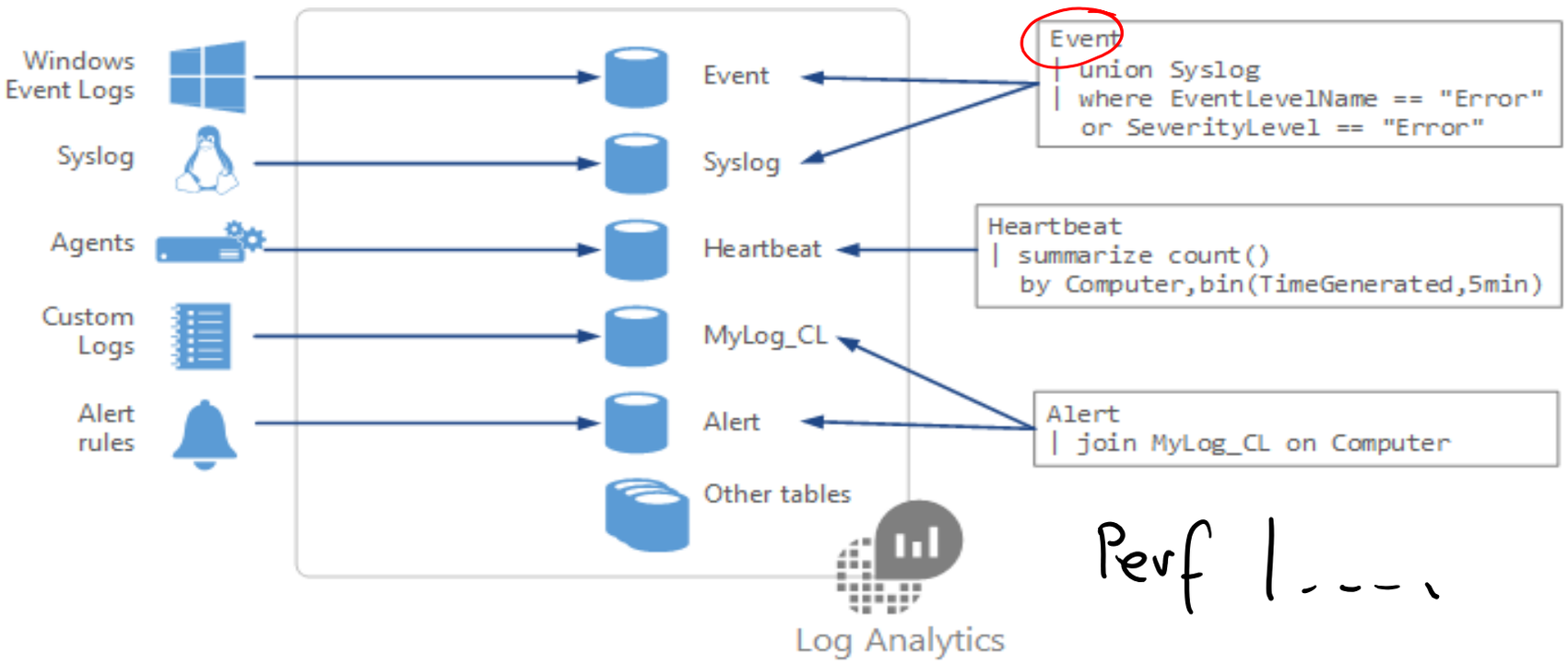
Consteaux Kusto Query Lang. KQL

Structure Log Analytics Queries



Perf |...

```
Event
| where (EventLevelName == "Error")
| where (TimeGenerated > ago(1days))
| summarize ErrorCount = count() by Computer
| top 10 by ErrorCount desc
```

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Lab – Implement Monitoring

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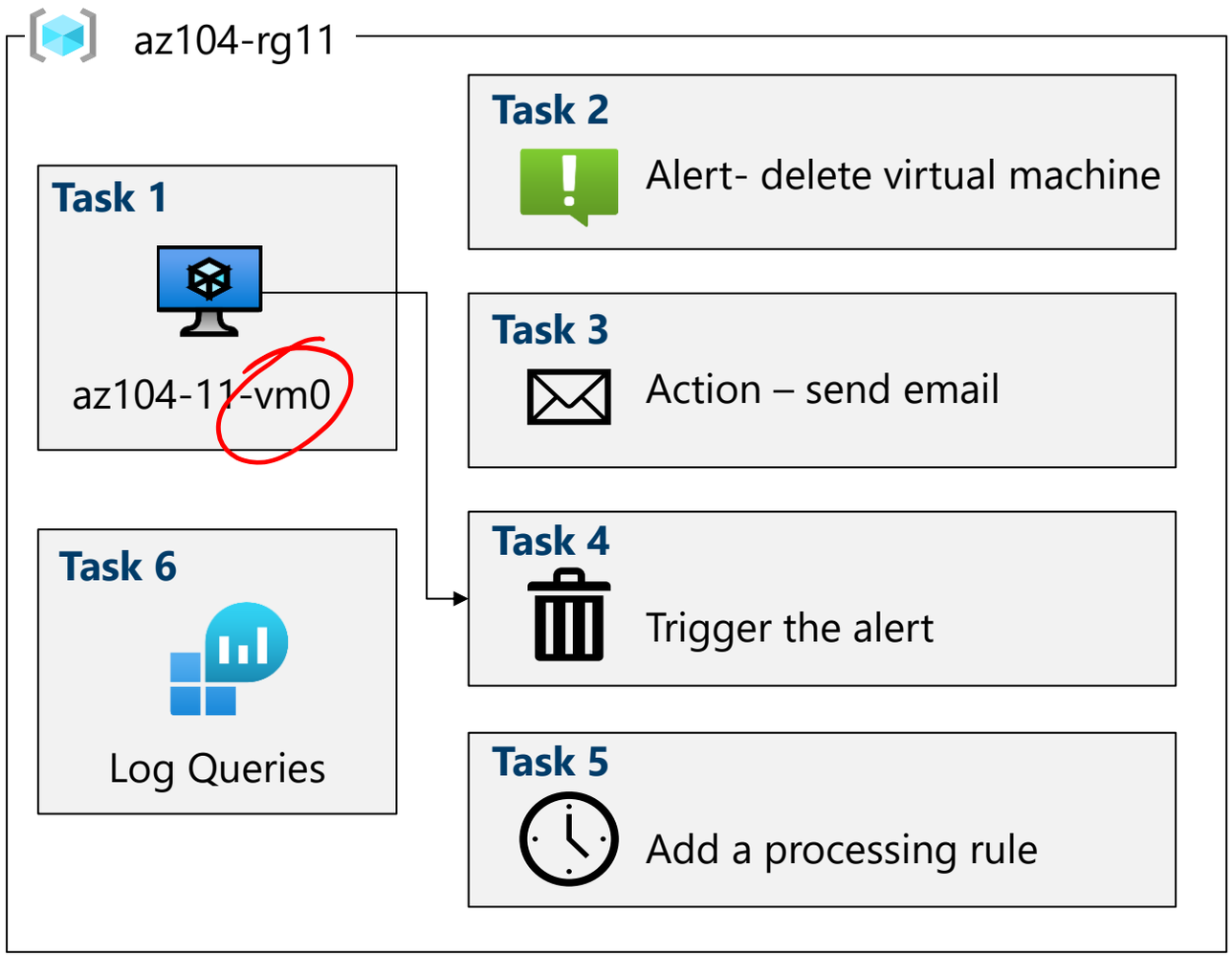
Lab 11 – Architecture diagram



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End of presentation

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